

RabbitMQ

- **open-source message broker** that enables **asynchronous** communication between microservices.
- helps decouple applications, so services **don't directly call** each other.
- uses **AMQP** (Advanced Message Queuing Protocol) — an application-layer protocol that transmits data in **binary frames**.

Key Concepts

- **Producer** → **Sends** messages
- **Consumer** → **Receives** messages
- **Queue** → **Stores** messages until consumed
- **Connection** → **TCP connection** between app and RabbitMQ
- **Channel** → Lightweight virtual connection over a TCP connection (used for publish/consume)
- **Exchange** → **Routes messages** to **queues** based on **rules**
- **Binding** → **Rule** that connects **exchange** to **queue**
- **Routing id** → Label used by **exchange** to decide **message routing**



Message Flow

- **Producer** publishes message to an **Exchange** via a **channel**.
- **Exchange** checks **bindings + routing key**.
- Message is routed to **matching queue(s)**.
- Message stays in **queue** until processed.
- **Consumer** receives and **acknowledges** it via a **channel**.

